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Course: Calculus Beta

Assignment: MyLab Math - Uge 3

Find the length of the curve $x = \frac{y^3}{9} + \frac{3}{4y}$ on $3 \leq y \leq 5$.

The length L of the curve on $a \leq y \leq b$ defined by $x = g(y)$ is $L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$.

The derivative with respect to y of $x = \frac{y^3}{9} + \frac{3}{4y}$ is $\frac{dx}{dy} = \frac{y^2}{3} - \frac{3}{4y^2}$.

Expand the square, and simplify.

$$\begin{aligned} 1 + \left(\frac{dx}{dy}\right)^2 &= 1 + \left(\frac{y^2}{3} - \frac{3}{4y^2}\right)^2 \\ &= 1 + \frac{y^4}{9} - \frac{1}{2} + \frac{9}{16y^4} \\ &= \frac{y^4}{9} + \frac{1}{2} + \frac{9}{16y^4} \end{aligned}$$

The expression $\frac{y^4}{9} + \frac{1}{2} + \frac{9}{16y^4}$ is a perfect square and is equal to $\left(\frac{y^2}{3} + \frac{3}{4y^2}\right)^2$.

Now, evaluate the integral to calculate the length.

$$\begin{aligned} L &= \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_3^5 \sqrt{\left(\frac{y^2}{3} + \frac{3}{4y^2}\right)^2} dy \\ &= \int_3^5 \left(\frac{y^2}{3} + \frac{3}{4y^2}\right) dy \\ &= \left[\frac{y^3}{9} - \frac{3}{4y}\right]_3^5 \\ &= \frac{(5)^3}{9} - \frac{3}{4(5)} - \left[\frac{(3)^3}{9} - \frac{3}{4(3)}\right] \\ &= \frac{989}{90} \end{aligned}$$

Simplify the radical.

Find the antiderivative.

Substitute the limits.

Simplify.

Thus, the length of the curve is $\frac{989}{90}$.